

Bucket Design for Continuous Timing Interfaces: Making “Continuous” Telemetry Auditably Discrete (without Redundancy)

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Abstract

Many anonymity mechanisms expose timing (request latency, retry spacing, refresh cadence). In this archive, deployable claims are expressed through an *equalization contract* on declared *interface projections* that can be audited from logs. Synthesis 25–26 provide conservative discrete log-audit certificates and anytime change-control for expanding audit menus. This note fills one narrow gap: how to turn a continuous timing signal into a *declared discrete projection* (a finite “bucket” alphabet) that is (i) compatible with the receipt schema, (ii) statistically audit-friendly, and (iii) non-redundant with the rest of the series.

1 Why buckets are the contract surface

Trace interfaces in this archive treat a projection as part of an exposure normal form (ENF): a deployment declares what is observed and what is forbidden/absent. [?] A continuous timing variable $T \in \mathbb{R}_{\geq 0}$ is generally *not* auditably equalized from finite logs without further structure. Accordingly, the recommended receipt-grade surface is a *bucket map*

$$b : \mathbb{R}_{\geq 0} \rightarrow \{1, \dots, m\}, \quad X := b(T),$$

and the equalization contract is stated for the discrete X (not the raw T). [?, ?]

Remark 1 (Scope). *If an adversary (or an internal observer) sees the raw T , then equalizing $X = b(T)$ alone does not control leakage from within-bucket variation. Receipt-grade claims should therefore be read as: the ENF forbids retaining or exposing finer timing information than X . This is the same “interface discipline” assumption used throughout the receipt series. [?, ?]*

2 Statistical effect of alphabet size

Let n be the number of samples available for a given class/policy k and menu item μ . Synthesis 25 provides a conservative radius bound for the multinomial empirical distribution on $\{1, \dots, m\}$. [?] Inverting the (approximate) dependence on m yields a simple design inequality.

Lemma 1 (Bucket budget from the Weissman tail bound). *Fix a per-item failure probability $\delta \in (0, 1)$ and a target TV radius $\tau \in (0, 1)$ for the empirical-to-true deviation $\text{TV}(\hat{P}, P) \leq \tau$ on an m -bucket alphabet. Under the Weissman-type bound used in Synthesis 25,*

$$\Pr[\text{TV}(\hat{P}, P) > \tau] \leq (2^m - 2) e^{-2n\tau^2},$$

so a sufficient condition for failure probability at most δ is

$$m \leq \log_2(2 + \delta e^{2n\tau^2}).$$

In the common “large- m ” regime where $2^m \gg 2$, this simplifies to the rule of thumb

$$m \lesssim \frac{2n\tau^2 - \log(1/\delta)}{\log 2},$$

which is meaningful only when the right-hand side is positive. Menu-level budgets δ_{stat} are allocated across items either by a static union bound (Synthesis 25) or an anytime spending schedule over audit events (Synthesis 26). [?]

3 Bucket design patterns

This note is intentionally conservative: its goal is a clean, publishable bucket declaration and a predictable audit cost.

Equal-width buckets (engineering-first). Choose boundaries $0 = t_0 < t_1 < \dots < t_m = \infty$ with fixed widths in a region of interest (e.g., milliseconds up to a deadline, then a catch-all tail bucket). This is simplest to implement and review, but can waste alphabet on rarely used regions.

Quantile buckets (audit-first). If a stable baseline exists (per class/policy), choose boundaries so that buckets have roughly equal mass under a reference distribution. This reduces sampling noise in thin tails. Because the boundaries depend on data, deployments should separate a *pilot/estimation* phase from the *frozen-audit* phase: publish the frozen boundaries in the receipt and treat boundary changes as a new menu item (or a new audit event) with its own spending. [?]

Deadline + tail-sign normal forms (mechanism-first). When the mechanism already enforces deadline-like behavior (padding, stop-time policies), choose buckets aligned to those normal forms: pre-deadline resolution, deadline bucket, and post-deadline tail buckets (often geometric in time). This keeps the bucket map consistent with the padding feasibility analyses in the published mathematics notes. [?, ?]

4 Receipt-facing declaration checklist

A bucketed timing projection should be publishable in a single place (ENF + receipt evidence), without repeating statistics:

- The bucket map b (explicit boundaries and the tail convention).
- The declared projection id (used by receipts to refer to the alphabet).
- A statement that finer timing telemetry is absent/forbidden by the ENF. [?]
- The log-audit certificate reference (Synthesis 25) and, if repeated, the spending rule reference (Synthesis 26). [?, ?]

References

- [1] Anonymous. Trace Interfaces for Anonymous DHT Deployments: Declared Projections and ENF Discipline. Synthesis 3, The-One-True-Archive (2026).

- [2] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Synthesis 12, The-One-True-Archive (2026).
- [3] Anonymous. Receipt Accounting Rulebook: How to Combine, Compare, and Replay Budget Line Items. Synthesis 19, The-One-True-Archive (2026).
- [4] Anonymous. Anonymous DHT Profiling and the Equalization Contract. Anonymity B, The-One-True-Archive (2026).
- [5] Anonymous. Discrete Equalization Audits from Logs: Menu-Level Finite-Sample TV Certificates. Synthesis 25, The-One-True-Archive (2026).
- [6] Anonymous. Anytime Statistical Spending for Menu Audits: Change-Control-Safe Confidence Budgets. Synthesis 26, The-One-True-Archive (2026).
- [7] Anonymous. Prefix Capacities and Separation Cuts for Shared-Kernel Termination-Time Hiding. Published mathematics note (2026-01-25), The-One-True-Archive.
- [8] Anonymous. Stop-Time Padding Addendum, Max-Mixture Separations and Tail-Sign Deadline Normal Forms. Published mathematics note (2026-01-26), The-One-True-Archive.